Algebraic Geometry Problem Set #3

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Problem (Exercise 1). *Let G be an algebraic group. Show that G is*

(a) separated,

(b) non-signular.

Hint for (a). Use the multiplication map $\mu : G \times G \to G$ to describe the diagonal.

Solution. Recall that G is an algebraic group if G is an algebraic variety with an identity element $e \in G$ and morphisms of algebraic varieties $\mu : G \times G \to G$ and $\iota : G \to G$ that satisfy the group axioms.

Part (a). To show that G is separated we need to show (by definition) that the diagonal $\Delta = \{(g, g) : g \in G\} \subseteq G \times G$ is a closed subset of $G \times G$. Now notice that

$$\Delta = \ker\Big(G \times G \to G : (g, h) \mapsto \mu(g, \iota(h))\Big).$$

Since kernels of the morphisms are closed, then so is Δ.

Part (b). For any $g \in G$, consider the morphism of varieties $\tau_g : G \to G$ given by $h \mapsto gh$. Note that this is an isomorphism since the inverse is given by $\tau_{g^{-1}}$. This isomorphism induces the isomorphism of local rings $\tau_g^{\#} : \mathcal{O}_{G,\tau(h)} \xrightarrow{\sim} \mathcal{O}_{G,h}$ for any point $h \in G$. **Conclusion:** all local rings $\mathcal{O}_{G,g}$ (for $g \in G$) are isomorphic.

Now recall that the set of non-singular points of G is a non-empty open subset. Thus, take some smooth point $s \in G$ and since any local ring is isomorphic to a *regular* local ring $\mathcal{O}_{G,s}$, we conclude that G is non-singular.

Problem (Exercise 2). *Let $G = \mathbb{G}_m^a$ and $H = \mathbb{G}_m^b$.*

(a) Prove that every map $f : G \to H$ with $f(1) = 1$ is a group homomorphism.

(b) Prove that group homomorphisms $G \to H$ form a group $\cong Hom(\mathbb{Z}^b, \mathbb{Z}^a)$.

Solution. Recall that $\mathbb{G}_m$ is the punctured affine line $\mathbb{A}^1 \setminus \{0\} = k^\times$ where the group structure is given by multiplication. Naturally, $\mathbb{G}_m^n$ is a set $(k^\times)^n$ where the multiplication is defined element-wise.

Part (a). Since the morphism f is defined by b separate coordinate functions: $f(x_1, \ldots, x_a) = (f_1(x), f_2(x), \ldots, f_b(x))$, it suffices to show that $\chi : G \to \mathbb{G}_m$ with $\chi(1) = 1$ is a homomorphism. Key observation we make:

$$\mathcal{O}(\mathbb{G}_m^n) \cong k[T_1^{\pm 1}, T_2^{\pm 1}, \ldots, T_n^{\pm 1}]. \quad \textit{(i.e., Laurent polynomials in n variables.)}$$

We finish off this part by presenting three claims (with *Claim #3* proving *part (a)*).

Claim #1 $\mathrm{Hom}_{\mathrm{Var}}(G, \mathbb{G}_m) \cong \mathcal{O}(G)^\times$ (in the category of varieties).

Claim #2 $\mathcal{O}_G^\times$ are precisely elements $\gamma \cdot T_1^{e_1} \ldots T_a^{e_a}$.

Claim #3 $\mathrm{Hom}_{\mathrm{AlgGrp}}(G, \mathbb{G}_m) \cong \mathbb{Z}^a$ (in the category of algebraic groups).

Proof of claims.

Claim #1. Any morphism of varieties $\varphi \in \mathrm{Hom}(G, \mathbb{G}_m)$ corresponds to the k-algebra homomorphism $\varphi^{\#} : \mathcal{O}(\mathbb{G}_m) \to \mathcal{O}(G)$ (or due to observation, equivalently,

$\varphi^{\#} : k[T^{\pm 1}] \to k[T_1^{\pm 1}, \ldots, T_a^{\pm 1}]$). This homomorphism is fully determined by where it sends T. Since $T \in \mathcal{O}(\mathbb{G}_m)^\times$ is invertible, $\varphi^{\#}(T) \in \mathcal{O}(G)^\times$. Therefore, picking φ boils down to picking element from $\mathcal{O}(G)^\times$.

Claim #2. This is a basic fact from the commutative algebra course.

Claim #3. Take a morphism of varieties $\chi : G \to \mathbb{G}_m$. From *Claim #1* and *Claim #2*, this morphism is of form $\chi(T_1, \ldots, T_a) = \gamma \cdot T_1^{e_1} \ldots T_a^{e_a}$ (for $\gamma \in k^\times$). This morphism is a homomorphism iff $\chi(x_1, \ldots, x_a)\chi(y_1, \ldots, y_a) = \chi(x_1 y_1, \ldots, x_a y_a)$ for every tuple $x, y \in G$. Plug in the expression of χ to get:

$$\chi(x_1 y_1, \ldots, x_a y_a) = \gamma x_1^{e_1} \ldots x_a^{e_a} y_1^{e_1} \ldots y_a^{e_a},$$
$$\chi(x_1, \ldots, x_a)\chi(y_1, \ldots, y_a) = \gamma^2 x_1^{e_1} \ldots x_a^{e_a} y_1^{e_1} \ldots y_a^{e_a}.$$

Thus $\chi \in \mathrm{Hom}_{\mathrm{AlgGrp}}(G, \mathbb{G}_m)$ iff $\gamma = 1$ (iff $\chi(1) = 1$). Hence, χ is determined by the exponents $(e_1, \ldots, e_a)$ picked from the lattice $\mathbb{Z}^a$.

Part (b). To give a map $G \to H$ means giving b group homomorphisms $G \to \mathbb{G}_m$, which from *part (a)* (*Claim #3*) boils down to picking the point on $\mathbb{Z}^a$. Therefore, we need to pick the matrix $\mathbb{Z}^{b \times a}$ which is equivalent to picking the morphism from $\mathrm{Hom}(\mathbb{Z}^b, \mathbb{Z}^a)$.

Problem (Exercise 3). *Let $Y \subseteq \mathbb{P}^2$ be a nonsingular projective plane curve defined by the homogeneous equation $f(x, y, z) = 0$ (for $f \in k[x, y, z]$ square-free, homogeneous of degree $d > 1$). Let $X = V(f) \subseteq \mathbb{A}^3$ be the "affine cone" of Y.*

(a) Show that the origin $0 \in \mathbb{A}^3$ is the unique singular point of X.

(b) Show that the strict transform $\widetilde{X}$ in the blowup of $\mathbb{A}^3$ at the origin is non-singular.

(c) Show that the tangent cone $\widetilde{X} \cap \pi^{-1}(0) \subseteq \mathbb{P}^2$ is equal to Y.

Solution. Further let $f(x, y, z) := \sum_{\alpha+\beta+\gamma=d} c_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma$.

Part (a). Suppose $(x_0, y_0, z_0) \in \mathbb{A}^3 \setminus \{0\}$ is a non-zero singular point of X. Since it is non-zero, it corresponds to the point in $\mathbb{P}^2$ which lies on Y. As all derivatives vanish, this is a singular point on Y, contradicting the fact that Y is nonsingular.

Finally, checking that $0 \in \mathbb{A}^3$ is indeed a singular point is straightforward.

Part (b) (here, I follow lecture notes, Section 6.4). Let the blowup of $\mathbb{A}^3$ be $\mathcal{B} = \{(a, x) \in \mathbb{P}^2 \times \mathbb{A}^3 : x \in L_a\}$ and let $\pi : \mathcal{B} \to \mathbb{A}^3$ be the projection map $(L, x) \mapsto x$. Note that since $\mathbb{P}^2$ is the union of three copies U_0, U_1, U_2 of the affine space $\mathbb{A}^3$, we can view $\mathcal{B}$ as the union of three open affine subvarieties $\mathcal{B}_i := \mathcal{B} \cap (U_i \times \mathbb{A}^3)$. Let coordinates of $\mathbb{A}^3$ be (x, y, z), while coordinates of $\mathbb{P}^2$ be $(U : V : W)$.

- The coordinates on $\mathcal{B}_0$ are (x, y, z) and $v = V/U$, $w = W/U$. The equation of $\mathcal{B}$ requires $xV = yU$, $xW = zU$, and $yW = zV$, which reduces to $xv = y$, $xw = z$, and $yw = zv$ (where the last one is redundant). Thus, our chart is given by $\mathcal{B}_0 = V(y - xv, z - xw) \subseteq \mathbb{A}^5$. Note that we have $\mathcal{B}_0 \cong \mathbb{A}^3$ with coordinates x, v, w and the map $\pi|_{\mathcal{B}_0}$ is the map $(x, v, w) \mapsto (x, vx, wx)$.
- $\mathcal{B}_1 \cong \mathbb{A}^3$ with coordinates $y, u = U/V, w = W/V$, with $x = uy$ and $z = wy$.
- $\mathcal{B}_2 \cong \mathbb{A}^3$ with coordinates $z, u = U/W, v = V/W$, with $x = uz$ and $y = vz$.

Recall that the strict transform $\widetilde{X} \subseteq \mathcal{B}$ is defined as

$$\widetilde{X} = \overline{\pi^{-1}(X \setminus \{0\})}.$$

Note that the origin of $\widetilde{X}$ is smooth iff it is smooth in every chart* $\mathcal{B}_0$, $\mathcal{B}_1$, and $\mathcal{B}_2$. Thus, let us check every chart.

Chart $\mathcal{B}_0$. Substitute $y := vx$ and $z := wx$, we get

$$\begin{aligned} f(x, vx, wx) &= \sum_{\alpha+\beta+\gamma=d} c_{\alpha\beta\gamma} x^\alpha (vx)^\beta (wx)^\gamma \\ &= \sum_{\alpha+\beta+\gamma=d} c_{\alpha\beta\gamma} x^{\alpha+\beta+\gamma} v^\beta w^\gamma \\ &= x^d \sum_{\alpha+\beta+\gamma=d} c_{\alpha\beta\gamma} v^\beta w^\gamma \\ &= x^d \widetilde{f}_0(v, w). \end{aligned}$$

Thus, $\widetilde{X} \cap \mathcal{B}_0$ is the curve $V(\widetilde{f}_0)$. But notice that $\widetilde{f}_0(v, w)$ is nothing but $f(1, v, w)$. Thus, if we had some singular point (x, v, w), then $(1 : v : w)$ would be a singular point of Y in $\mathbb{P}^2$, which is impossible by definition of Y.

Chart $\mathcal{B}_1$. Here, $\widetilde{X} \cap \mathcal{B}_1 \cong V(\widetilde{f}_1)$ with $\widetilde{f}_1 = f(u, 1, w)$. Similarly we do not have any singularities here.

Chart $\mathcal{B}_2$. Here, $\widetilde{X} \cap \mathcal{B}_2 \cong V(\widetilde{f}_2)$ with $\widetilde{f}_2 = f(u, v, 1)$. No singularities here either.

Thus we conclude that $\widetilde{X}$ is smooth.

Part (c). To find the tangent cone, we need to intersect each chart with $\pi^{-1}(0)$ restricted to this chart. Specifically, it means that in $\mathcal{B}_0$ we set $x = 0$, in $\mathcal{B}_1$ we set $y = 0$, while in $\mathcal{B}_2$ we set $z = 0$. But note that corresponding coordinates do not appear in the expressions of $\widetilde{f}_0$, $\widetilde{f}_1$, and $\widetilde{f}_2$, respectively! Thus:

$$\begin{aligned} \widetilde{X} \cap \mathcal{B}_0 &= \{(1 : v : w) : f(1, v, w) = 0\}, \\ \widetilde{X} \cap \mathcal{B}_1 &= \{(u : 1 : w) : f(u, 1, w) = 0\}, \\ \widetilde{X} \cap \mathcal{B}_2 &= \{(u : v : 1) : f(u, v, 1) = 0\}. \end{aligned}$$

Hence, the union of these three patches is exactly $Y = V(f) \subseteq \mathbb{P}^2$.

Problem (Exercise 4). *A sheaf $\mathcal{F}$ on a topological space X is called flasque (aka flabby) if for every inclusion $V \subseteq U$ of open sets, the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective.*

(a) Show that a constant sheaf on an irreducible topological space is flasque.

(b) Suppose that $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is an exact sequence of sheaves and that $\mathcal{F}'$ is flasque. Show that for every open $U \subseteq X$, the sequence $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U) \to 0$ is exact.

(c) Let $f : Y \to X$ be a continuous map and let $\mathcal{F}$ be a flasque sheaf on Y. Show that the sheaf $f_*\mathcal{F}$ on X is flasque.

(d) For any sheaf $\mathcal{F}$, the sheaf $D(\mathcal{F})$ of discontinuous sections of $\mathcal{F}$ is flasque.

Solution.

Part (a). For an irreducible topological space, the global section of the constant

*To be honest, I do not quite understand how to show this formally.

sheaf k_X is given by:

$$k_X(U) = \begin{cases} k, & \text{if } U \neq \emptyset \\ \{0\}, & \text{if } U = \emptyset. \end{cases}$$

Thus, we have three cases:

- Case #1. $V = \emptyset$, U is any open. Then, $k \to \{0\}$ is clearly surjective.
- Case #2. V, U any non-empty opens. Then, the restriction map $k \to k$ is an identity map id_k, which is surjective.
- Case #3. $U = \emptyset$, $V = \emptyset$. Then, the map $\{0\} \to \{0\}$ is surjective as well.

Part (b). Recall that the global section function $\Gamma(\cdot, \mathcal{F})$ is left-exact, so we already know that $0 \xrightarrow{\alpha} \mathcal{F}'(U) \xrightarrow{\beta} \mathcal{F}(U) \to \mathcal{F}''(U)$ is exact. So the real question of this part is to show that the map $\mathcal{F}(U) \to \mathcal{F}''(U)$ is surjective given $\mathcal{F}'$ is flasque.

Take some section $s'' \in \mathcal{F}''(U)$. We want to construct $s \in \mathcal{F}(U)$ whose image in $\mathcal{F}''(U)$ is s''. Since taking stalk $\cdot_x$ is an exact functor (for any $x \in U$), there exists an open neighbourhood $V_x \subseteq U$ and $s_x \in \mathcal{F}(V_x)$ with $s_x \mapsto s''|_{V_x}$. So *locally* s'' lifts to sections of $\mathcal{F}$, so now we need to somehow glue local sections into a global section of U using the flasqueness of $\mathcal{F}'$.

That said, consider the open convering $\{V_x\}_{x \in U}$ of U with corresponding "lifted" local sections $\{s_x \in \mathcal{F}(V_x)\}_{x \in U}$. On overlaps $V_x \cap V_y \neq \emptyset$, the differences of sections $s_x - s_y \in \ker(\mathcal{F}(V_x \cap V_y) \to \mathcal{F}''(V_x \cap V_y))$ (since the images of s_x and s_y agree on $V_x \cap V_y$). Using exactness of the sequence of sheaves, this implies there exists some $s'_{xy} \in \mathcal{F}'(V_x \cap V_y)$ such that $r_{x,y} \mapsto (s_x - s_y)|_{V_x \cap V_y}$. But since $\mathcal{F}'$ is flasque, restriction maps $\mathcal{F}'(V_x) \to \mathcal{F}'(V_x \cap V_y)$ are surjective. Thus, we can lift s'_{xy} up to some $\widetilde{s}'_{xy} \in \mathcal{F}'(V_x)$ such that $\widetilde{s}'_{xy}|_{V_x \cap V_y} = s_{xy}$.

Finally, we define $\widetilde{s}_x := s_x - \alpha(\widetilde{s}'_{xy})$. Then, one checks $\widetilde{s}_x|_{V_x \cap V_y} = \widetilde{s}_y|_{V_x \cap V_y}$, so we can glue them together to get $s \in \mathcal{F}(U)$ with $s|_{V_x} = \widetilde{s}_x$. Finally, s maps precisely to s'' by the map $\mathcal{F}(U) \to \mathcal{F}''(U)$, as was required.

Part (c). Consider any two opens $V \subseteq U \subseteq X$. Since f is continuous, $f^{-1}(V) \subseteq f^{-1}(U)$, thus $\mathcal{F}(f^{-1}(U)) \to \mathcal{F}(f^{-1}(V))$ is surjective since $\mathcal{F}$ is flasque. However, this is exactly the map $f_*\mathcal{F}(U) \to f_*\mathcal{F}(V)$, which we've shown is surjective.

Part (d). Recall that $D(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x$. Note that for $V \subseteq U$, the map $\prod_{x \in U} \mathcal{F}_x \to \prod_{x \in V} \mathcal{F}_x$ is simply the projection (i.e., we forget coordinates outside V). Therefore, take any element $(s_x)_{x \in V} \in \prod_{x \in V} \mathcal{F}_x$ and define the inverse by keeping the same s_x for all $x \in V$, while for $U \setminus V$ choose your favourite germ :)

Problem (Exercise 5). *Let X be an algebraic set and let $\mathcal{O}_X$ be its sheaf of regular functions. Let $Y \subseteq X$ be a closed subset and let $i : Y \hookrightarrow X$ be the inclusion.*

(a) For each open subset $U \subseteq X$, let $\mathcal{I}_Y(U)$ be the ideal in the ring $\mathcal{O}_X(U)$ consisting of those regular functions which vanish at all points of $Y \cap U$. Show that the presheaf $U \mapsto \mathcal{I}_Y(U)$ is a sheaf. It is called the sheaf of ideals $\mathcal{I}_Y$ of Y, and it is a subsheaf of the sheaf of rings $\mathcal{O}_X$.

(b) Show that the quotient sheaf $\mathcal{O}_X/\mathcal{I}_Y$ is isomorphic to $i_*\mathcal{O}_Y$, so that we have a short exact sequence $0 \to \mathcal{I}_Y \to \mathcal{O}_X \to i_*\mathcal{O}_Y \to 0$.

(c) Consider the case $X = \mathbb{P}^1$ and $Y = \{x, y\}$ for distinct points $x, y \in X$. Show that $\mathcal{O}_X(X) \to (i_*\mathcal{O}_Y)(X)$ is not surjective.

Solution.

Part (a). For $V \subseteq U$, the restriction maps $\mathcal{I}_Y(U) \to \mathcal{I}_Y(V)$ are still the usual restrictions $s \mapsto s|_V$ (note that if s vanishes on $Y \cap U$, it definitely vanishes on the subset $Y \cap V \subseteq Y \cap U$).

Now we prove the gluing axiom. Assume that $\{U_\alpha\}$ is an open cover of any open U and $\{s_\alpha \in \mathcal{I}_Y(U_\alpha)\}$ satisfy $s_\alpha|_{U_\alpha \cap U_\beta} = s_\beta|_{U_\alpha \cap U_\beta}$ in $\mathcal{I}_Y(U_\alpha \cap U_\beta)$. Now, invoking the gluing axiom of sheaf $\mathcal{O}_X$, we construct $s \in \mathcal{O}_X(U)$ such that $s|_{U_\alpha} = s_\alpha$. Since each s_α vanishes on $Y \cap U_\alpha$, s vanishes on $\bigcup_\alpha (Y \cap U_\alpha) = Y \cap U$, hence $f \in \mathcal{I}_Y(U)$.

Part (b). By definition, $(i_*\mathcal{O}_Y)(U) = \mathcal{O}_Y(U \cap Y)$. Thus, the map $\varphi : \mathcal{O}_X \to i_*\mathcal{O}_Y$ sends $s \in \mathcal{O}_X(U)$ to $s|_{U \cap Y} \in \mathcal{O}_X(U \cap Y)$. By definition, $\ker(\mathcal{O}_X \to i_*\mathcal{O}_Y)$ are those regular functions that vanish on the intersection $U \cap Y$, which is exactly $\mathcal{I}_Y(U)$. Therefore, we already know exactness of $0 \to \mathcal{I}_Y \to \mathcal{O}_X \xrightarrow{\varphi} i_*\mathcal{O}_Y$. What remains to show is that φ is surjective. It suffices to do this at some stalk $y \in Y$ (for $y \in X \setminus Y$ the stalk $(i_*\mathcal{O}_Y)_y$ is trivially zero, so the map is surjective). Notice that $(i_*\mathcal{O}_Y)_y \cong \mathcal{O}_{Y,y}$. Indeed,

$$(i_*\mathcal{O}_Y)_y = \varinjlim_{U \ni y} (i_*\mathcal{O}_Y)(U) \cong \varinjlim_{U \ni y} \mathcal{O}_Y(U \cap Y) \cong \varinjlim_{V \ni y} \mathcal{O}_Y(V) = \mathcal{O}_{Y,y}.$$

Thus, we reduced the problem to considering the natural restriction map on germs $\mathcal{O}_{X,y} \to \mathcal{O}_{Y,y}$, but this map in turn is clearly surjective.

The isomorphism thus follows from the exactness of the sequence.

Part (c). Recall that $\mathcal{O}_X(X) \cong k$ (global regular functions on a projective variety are constants) while $(i_*\mathcal{O}_Y)(X) = \mathcal{O}_Y(Y) \cong k \times k$ (regular functions on two points is simply choosing a value in k for x and y). Thus, the map $\mathcal{O}_X(X) \to (i_*\mathcal{O}_Y)(X)$ is simply a map $k \to k \times k$ given by $\gamma \mapsto (\gamma, \gamma)$, which is not surjective.